Introduction to AI and Data Science

Week 3

Lab Activities

Activity 1: Build an expert system

In groups, complete a knowledge base of things each of you like. The goal is to develop an expert system that can identify students based on a list of facts of what the student likes. Start with

```
rules = {
    "Nick": [["Mathematics", "Sushi", "The Manic Street Preachers"]],
    "Sushi":[["Japanese food"]],
    "Katsu curry":[["Japanese food"]]
}
```

Add enough rules so that each person in your group can be uniquely identified by their likes.

Add some more rules like the "Sushi":[["Japanese food"]] rule, so that we can ask more broad questions of our expert system.

Activity 2: Build a rules-based expert system

In groups, complete a knowledge base of animal facts. The goal is to develop an expert system that can identify animals based on a list of facts. Start with

```
rules ={
  "Bird": [["lays eggs", "flies"], ["has feathers"]],
  "Penguin": [["is aquatic", "Bird"]],
  "Otter": [["is aquatic", "has fur"]]
}
```

Add enough rules so that several different animals can be uniquely identified from observable facts.

Activity 3: Build a semantic net

As a whole class (or in large groups) complete a knowledge base of things each of you like. The goal is to develop a semantic net that gives us a rich picture of who is in our class.

```
student_net = [
    ("Nick", "likes", "Mathematics"),
    ("Nick", "likes", "Sushi"),
    ("Nick", "likes", "The Manic Street Preachers"),
    ("Sushi", "isA", "Japanese Food"),
```

("Mathematics", "isA", "Academic Subject"),

("The Manic Street Preachers", "isA", "Rock Band"),

("The Manic Street Preachers", "hasNationality", "Welsh")

]

Add enough triples so that a variety of questions can be answered about the group.

Activity 4: Can You Diagnose the Patient?

A simple expert system uses these rules:

Rule	IF	THEN
R1	respiratory issue **AND** fever	pneumonia
R2	cough	respiratory issue
R3	wheezing	respiratory issue
R4	fever **AND** spots	measles

Patient symptoms

The patient currently has:

- cough
- fever
- spots

So our starting facts are:

{cough, fever, spots}

Your task

Use **forward chaining** to find out what else the system can conclude.

Work through the rules **from R1 to R4**.

For each rule, ask:

“Do I currently have all the facts needed for this rule?”

- If **yes** → fire the rule and add the new fact.
- If **no** → move to the next rule.
- Once a rule has fired, do not fire it again.
- Start another cycle if new facts were added.

Complete the table:

Cycle	Rule	Can it fire?	New fact added
1	R1	?	?
1	R2	?	?
1	R3	?	?
1	R4	?	?
2	R1	?	?

Final questions

1. What new facts did you discover?
2. What diagnosis or diagnoses did the system reach?
3. Why could the system not diagnose pneumonia at the very beginning?
4. When should the forward-chaining process stop?

Hint

A new fact discovered in one cycle can help another rule fire in the next cycle.

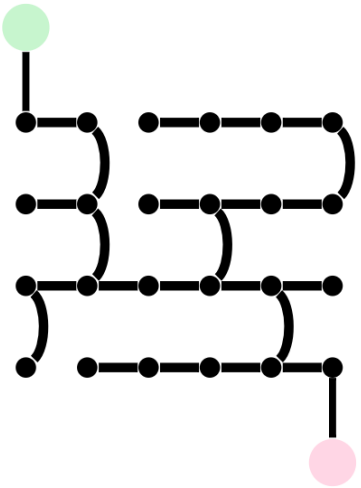
For example:

cough → respiratory issue

and then:

respiratory issue + fever → ?

Activity 5: Depth-First Search for a maze



Apply the Depth-First Search to escape this maze (start at the green node on the top, and finish at the red node on the bottom).

You will need to

- label the nodes
- choose an order to add the unvisisted neighours.

Write down the **current_node**, **visited list** and **stack** at each step.

Activity 6: A game with an infinite branch

Consider a simple game where:

- the player starts with no coins.
- the player is trying to buy a sandwich.

On each turn, the player can either

- buy a sandwich that costs 5 coins, or
- collect another coin.

What are the states for this game?

What possible results could a depth-first search give?

Use this example to write about depth-first search using one or more of the evaluation terms above.